Lecture VI – Neural Network

Neural Network:

Neural networks are modeled on the human brain. Just as neurons are networked to one another, artificial neural networks emphasize the relationships between attributes. They have proven especially effective in analyzing visual, audio, and written language data. Similar to Principle Component Analysis, they are affective at identifying which features are most relevant.

Perceptron

The Perceptron algorithm attempts to classify objects using a single line, plane, or hyperplane. It begins with a set of weights matching the number of dimensions under consideration plus 1 additional weight. For example, if we're looking at a two dimensional field, we will have three weights, ω_0, ω_1, and ω_2. These values will be multiplied by the value of their corresponding dimensions and added together, this is ω Transpose x:

$$\omega^T x = \omega_0 * x_0 + \omega_1 * x_1 + \omega_2 * x_2$$

Assuming a threshold of 0, if the result is greater than 0, the object is *activated*, similar to a neuron being activated in the brain. *Activated* means the object will be classified as label 1. If it is negative, it will be classified as label 0.

The ω vector and x vector can be noted like this:

$$\begin{bmatrix} \omega_0 \\ \omega_1 \\ \omega_2 \end{bmatrix} \qquad \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$$

Example: an object with x, y values of (-3, 5), and the randomly selected weights of 0, 1, .5, the vectors would be:

$$\begin{bmatrix} 0 \\ 1 \\ .5 \end{bmatrix} \qquad \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}$$

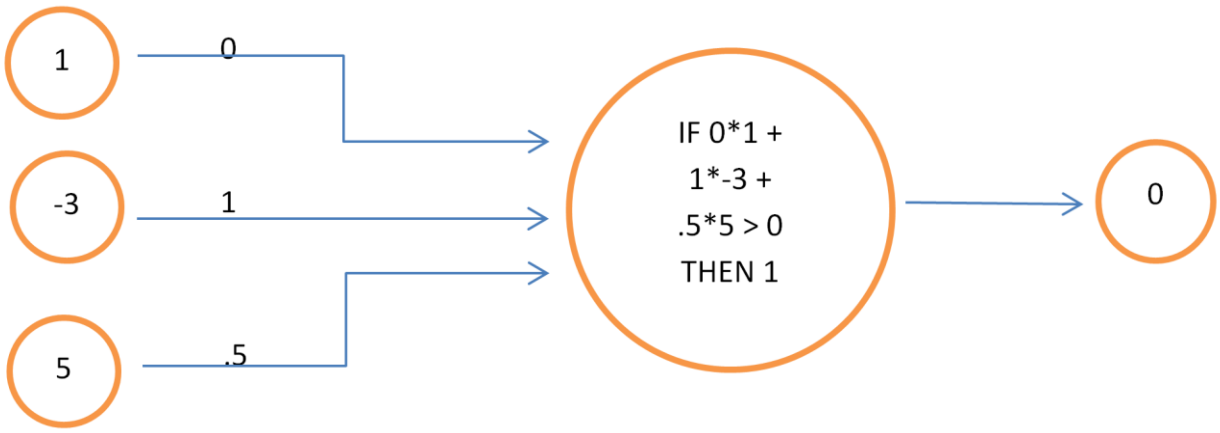
Note:

In the x vector, $x_2 = y$, $x_1 = x$, and $x_0 = 1$. X_0 will always equal 1. This will allow the first weight, ω_0 or *bias*, to pass through.

And the equation for ω T x would be:

0*1 + 1*-3 + .5*5 = -.5

A visual perspective:



A Linear perspective:

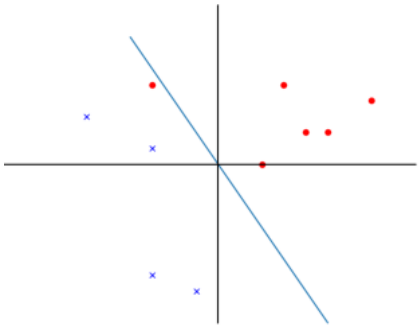
Leaving x and y as variables in the equation above gives us:

0*1 + 1x + .5y

When simplified, this is the equation of a line:

Y = -2x

Which would look like this:



The object in this example is the only red dot that appears below the blue line. If we calculated ω T x for every other object, we would notice that all the red dots will be classified as 1's (above the line) and all the blue x's would be classified as 0 (below the line).

Learning

In the example above, the object will be classified as 0 because -.5 is less than 0. However, because this is supervised learning, we know the class should be 1. The model needs to *learn* how to correct for this mistake.

To correct the misclassification, we will adjust the weights. We will adjust the weights by a parameter called the learning rate, or λ. If the ω T x needs to go down, λ will be negative, otherwise it will be positive. Each ω value will be adjusted according to this equation:

$\omega' = \omega + \lambda dx$

Where ω' is the new weight, ω is the old weight, and d is -1 if the ω T x needs to go down or 1 if it needs to go up.

For example, to classify the object above as 1, with a λ of .2, the equation for each new weight would be:

.2 = 0 + (.2)(1)(1)

.4 = 1 + (.2)(1)(-3)

1.5 = .5 + (.2)(1)(5)

Note: d = 1 because the object needs to go up.

The new ω and x vectors would be:

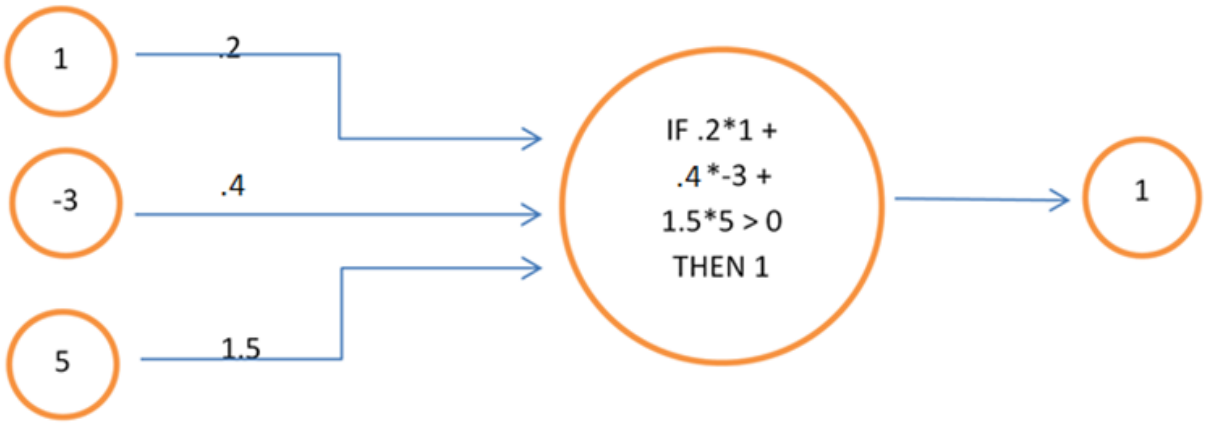
$$\begin{bmatrix} .2 \\ .4 \\ 1.5 \end{bmatrix} \qquad \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}$$

Now, the equation for ω' T x would be:

$$.2*1 + .4*-3 + 1.5*5 = 6.5$$

Because 6.5 is greater than 0, the object is classified by 1 after the weight change.

Visual perspective:



Linear Perspective:

The new equation for the line represented by the ω vector is:

$$y = -(.4x + .2)/1.5$$

